

Relation Algebra

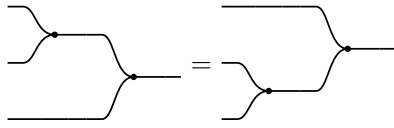
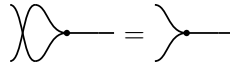
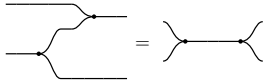
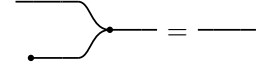
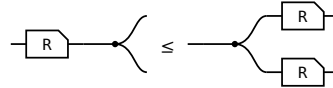
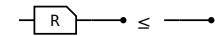
§2. Relations

(2a)

$\mathbf{Rel}$ is a poset-enriched category with $(\otimes, \circ)$ where $\otimes$ is commutative cartesian product, and converse $\circ \dashv \circ$, and

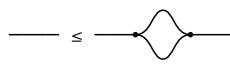
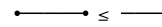
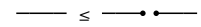
$C: (\triangleright : A \otimes A \rightarrow A, \multimap : I \rightarrow A)$ a commutative monoid with $C^\circ \dashv C$. We write C° as $(\triangleleft : A \rightarrow A \otimes A, \multimap : A \rightarrow I)$

(2b)

 $\triangleright$ associative $\triangleright$ commutative $\circ$ is its unitFrobenius, one half — the other is its $\circ$ $R \triangleleft \leq \triangleleft (R \otimes R)$ $R \multimap \leq \multimap$

§2.1. $\forall$ object A , $(A, \triangleleft, \multimap) \dashv (A, \triangleright, \multimap)$

(2.1a)

 $\triangleright \triangleleft \leq 1$  $1 \leq \triangleleft \triangleright$  $\circ \multimap \leq 1$  $1 \leq \multimap \circ$

The last of these is the only one that makes a picture **bigger**, and it is worth a name: **a wire is below the cut wire**. In $\mathbf{Rel}$ it reads $\{(a, a)\} \subseteq A \times A$ — cut a wire and its two ends stop having to agree, so cutting can only add pairs. It is the one weakening this calculus gives away for free.